Hallucination and Knowledge-Cutoff Probe for a Library Assistant

Abstract

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Large language models (LLMs) are increasingly used as conversational library assistants that answer questions about books, authors, availability and library policies. Two weaknesses limit their trustworthiness: **hallucination**, in which the model produces fluent but fabricated titles, identifiers, citations or rules; and the **knowledge cutoff**, in which the model is unaware of anything after its training date, such as new acquisitions, new editions or revised policies. This project develops a **Hallucination and Knowledge-Cutoff Probe**: a structured set of test queries, built from the library's own catalogue, that measures how often and in what way an assistant hallucinates and how well it recognises the limits of its knowledge. The probe comprises a five-type hallucination taxonomy, tiered catalogue-grounded, adversarial and unanswerable queries, four temporal probe families, and metrics including per-type hallucination rate, temporal staleness rate, cutoff awareness score and deferral precision. The findings drive a mitigation layer of retrieval grounding, identifier copying, claim verification and abstention. The expected outcome is a reproducible evaluation procedure that lets a library choose, monitor and safely deploy an LLM assistant.

Keywords: hallucination, knowledge cutoff, library assistant, LLM evaluation, retrieval-augmented generation